Home

Dr. Shirley Jackson

We certify that the content included on this website is original work (or is cited accurately) and accurate to our knowledge. Work included which was conducted as part of a team is indicated and attributed.

Matthew Ritten, Jay Duggal, and Youssef Ahmar

Layout by Bootstrap.

Our Innovator

Brief Background

Dr. Shirley Jackson,was born onAugust 5th , 1946. During her childhood, she took a particular interest in science and would often do science experiments. She would go on to be the first black woman to earn a PhD in physics from MIT and would follow that up by becoming a leading expert in theoretical physics. She would go onto work at Bell Laboratories, “her research there contributed to the inventions of the touch-tone telephone, fiber-optic cables, caller ID, and call waiting.” (Ramanathan, 2023). Following this, in 1995, she was appointed the head of the Nuclear Regulatory Commission, and then, finally, “she became the eighteenth president of Rensselaer Polytechnic Institute.” (The HistoryMakers, 2006).

Under-Representation

Dr. Shirley Jacksonis one of strongest examples of underrepresentation, in MIT she was, “among the first African American students to attend MIT, and in her undergraduate class she was one of only two women.” (The HistoryMakers, 2006).To follow that up she would go on to be the first woman to earn a PhD in physics from MIT. Beyond that, she excelled in an environment that was institutionally hostile against her, as she was forced to attend segregated schools growing up and was told by her school advisor that “colored girls should learn a trade.” (Success Academy 2019). She overcame these structural hurdles by believing in herself and pushing towards her goals no matter what. By now it is pretty safe to say that Dr. Shirley Jackson, who was the head of The U.S. Nuclear Regulatory Commission, Principal of the Rensselaer Polytechnic Institute, the co-chair of the President’s Intelligence Advisory Board, and recipient of the National Medal of Science, has overcome the structural prejudice against her and excelled despite her circumstances.

Contribution

WhileworkingatBell Laboratories, “her research there contributed to the inventions of the touch-tone telephone, fiber-optic cables, caller ID, and call waiting.” (Ramanathan, 2023).

Future Promise

Earlyresearch by Dr. Shirley Ann Jackson in telecommunication enabled modern forms of communications like fiber optic and caller ID, which also help define future technologies today. Her research laid the groundwork for future advances in 6G networks, which will enable faster global communication, virtual reality, and smart city networks. Her physics training also ties into the growing field of quantum communication, which provides ultra-secure transmission of information. As technology progresses, the legacy of Jackson will keep setting the pace for a faster, smarter, and more accessible world to connect worldwide.

Selection

Our decision wasbetween Hedy Lamar and Dr. Shirley Jackson, both were innovators who experienced underrepresentation and had to overcome the odds to have been incredibly influential on technology. For context Hedy Lamar during World War 2, used her passion for science to help her invent a frequency-hopping signal that prevents torpedoes from being jammed or tracked. The tech she invented is still used in Bluetooth and Wi-Fi, being an important component to it. However we decided against her being our innovator for two main reasons, one we felt that she would be a common pick. Secondly, when we looked up Dr. Shirley Jackson, a different person came up, this solidified in our minds that Dr. Shirley Jackson was truly underrepresented.

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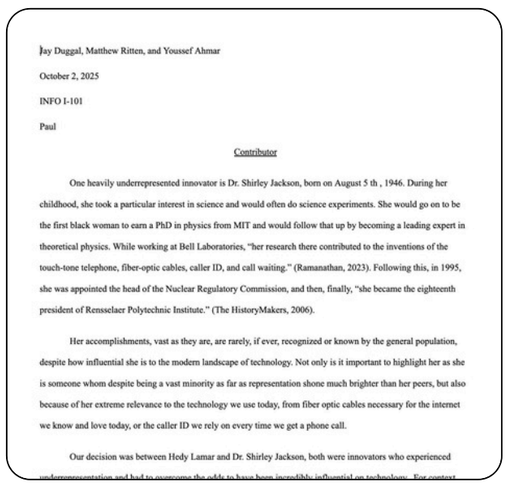
Our Process



Jay Duggal, Matthew Ritten, and Youssef Ahmar

October 2, 2025

INFO I-101

Paul

Contributor

One heavily underrepresented innovator is Dr. Shirley Jackson, born on August 5 th , 1946. During her childhood, she took a particular interest in science and would often do science experiments. She would go on to be the first black woman to earn a PhD in physics from MIT and would follow that up by becoming a leading expert in theoretical physics. While working at Bell Laboratories, "her research there contributed to the inventions of the touch-tone telephone, fiber-optic cables, caller ID, and call waiting." (Ramanathan, 2023). Following this, in 1995, she was appointed the head of the Nuclear Regulatory Commission, and then, finally, "she became the eighteenth president of Rensselaer Polytechnic Institute." (The HistoryMakers, 2006).

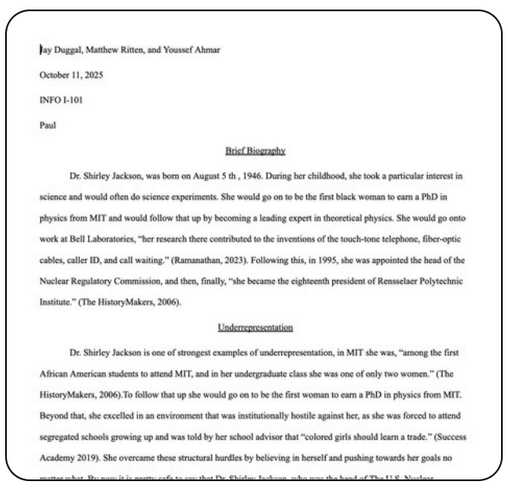
Her accomplishments, vast as they are, are rarely, if ever, recognized or known by the general population, despite how influential she is to the modern landscape of technology. Not only is it important to highlight her as she is someone whom despite being a vast minority as far as representation shone much brighter than her peers, but also because of her extreme relevance to the technology we use today, from fiber optic cables necessary for the internet we know and love today, or the caller ID we rely on every time we get a phone call.

Our decision was between Hedy Lamar and Dr. Shirley Jackson, both were innovators who experienced

Project Charter

Thepurpose of thischarter is to indicate our group's choice for the focus of our project and to outline a general plan of attack our group can follow throughout the semester. The charter is a brief 3-4 page document that identifies what underrepresented person in informatics we would like to highlight and our overall group plan for completing the project.

Click here to look at the Google Doc!



Jay Duggal, Matthew Ritten, and Youssef Ahmar

October 11, 2025

INFO I-101

Paul

Brief Biography

Dr. Shirley Jackson, was born on August 5 th , 1946. During her childhood, she took a particular interest in science and would often do science experiments. She would go on to be the first black woman to earn a PhD in physics from MIT and would follow that up by becoming a leading expert in theoretical physics. She would go onto work at Bell Laboratories, "her research there contributed to the inventions of the touch-tone telephone, fiber-optic cables, caller ID, and call waiting." (Ramanathan, 2023). Following this, in 1995, she was appointed the head of the Nuclear Regulatory Commission, and then, finally, "she became the eighteenth president of Rensselaer Polytechnic Institute." (The HistoryMakers, 2006).

Underrepresentation

Dr. Shirley Jackson is one of strongest examples of underrepresentation, in MIT she was, "among the first African American students to attend MIT, and in her undergraduate class she was one of only two women." (The HistoryMakers, 2006).To follow that up she would go on to be the first woman to earn a PhD in physics from MIT. Beyond that, she excelled in an environment that was institutionally hostile against her, as she was forced to attend segregated schools growing up and was told by her school advisor that "colored girls should learn a trade." (Success Academy 2019). She overcame these structural hurdles by believing in herself and pushing towards her goals no

Project Research

The purpose of thisdocumentisto assemble research on our innovator, Dr. Shirley Jackson. It shows her technological contributions, and the future of innovation based on her contributions. It is a full, complete profile for Dr. Shirley Jackson.

Click here to look at the Google Doc!

Project Website Draft

Project Website Draft

Project Data Viz

Project Data Viz

Project Final Website

Project Final Website

Project Final Submission

Project Final Submission

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Our Team

Jay Duggal

Jayisadetermined, creative, and knowledge-hungry professional currently based at Indiana University Bloomington, where he is a cybersecurity major and a sophomore. With a solid educational foundation and a passion for growing both personally and professionally, he brings enthusiasm and adaptability to every opportunity. Jay's drive to learn and innovate makes him an asset in dynamic environments where fresh ideas and dedication matter.

Connect on LinkedIn!

Youssef Ahmar

Youssef isahardworking studentatIndiana University Bloomington where he is majoring in Public Financial Management through the O'Neill school of public affairs. Youssef uses his strong work ethic, enthusiasm for learning, and personal development skills to achieve goals and create opportunities.

Connect on LinkedIn!

Matthew Ritten

Matthew is a determined freshman atIndiana University in Bloomington, Indiana, studying Informatics and minoring in Data Science. With an array of interests from programming, sound design, and several others, Matthew continually puts his best effort into what he wants and believes in. Matthew's leadership and ability to take initiative allows him to be dynamic in any group he particpates in. With experience as a team member in a sport, at jobs, and in school; solid foundational education; and a drive to push himself and others to their best; Matthew is eager to show the world what he is capable of.

Connect on LinkedIn!

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